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Epilogue

Congratulations on reaching *the final chapter!*

This is the last stop in our journey. Before we close, we'll do a number of things:

- Summarize what we've learned in the book
- Discuss five steps to get the best out of your causal projects
- Take a look at the intersection of causality and business and see how organizations implement successful causal projects
- Take a sneak peek into (a potential) future of causality
- Discuss where to find resources and how to learn more about causality

Ready for the last leap?

What we've learned in this book

Back in *Chapter 1*, we started our causal journey by asking about the reasons to use causal modeling rather than traditional machine learning, despite the tremendous success of the latter.

We defined the concept of confounding and showed how it can lead us astray by producing spurious relationships between causally independent variables. Next, we introduced **the Ladder of Causation** and its three rungs – **observations**, **interventions**, and **counterfactuals**. We showed the differences between observational and interventional distributions using linear regression.

After that, we refreshed our knowledge of the basic graph theory and introduced graphs as an important building block for causal models. We discussed three basic conditional independence structures – **forks**, **chains**, and **colliders**, and showed that colliders have a special status among the three, allowing us to infer the direction of causal influence from the data.

Next, we introduced **DoWhy**, and we learned how to implement DoWhy's four-step causal process. This led us to the discussion on assumptions behind causal models and the challenges that we might face when implementing them in practice.

Keeping these ideas in mind, we moved to **(conditional) average treatment effect ((C)ATE)** estimation methods and demonstrated how to implement them using DoWhy and EconML in observational and interventional scenarios.

We closed *Part 2* of the book by implementing meta-learners with tree-based and deep learning methods, and we learned how to implement a Transformer-based CausalBert model that allows us to control for confounding coming from natural language.

In *Part 3*, we reviewed various sources of causal knowledge and headed toward a discussion on causal discovery. We started with classic constraint-based algorithms and moved toward modern gradient-based methods, closing with a full implementation of Microsoft's DECI algorithm using PyTorch.

In the next section, we'll discuss five steps to get the best out of your causal project that are based on some of the best practices we discussed in the book.

Five steps to get the best out of your causal project

In this section, we'll discuss five steps that can help you maximize the potential of your causal project.

Starting with a question

Starting with a well-defined question is a necessary step of any scientific or business endeavor, but it has special importance in causality.

A well-defined question can transform an impossible problem into a tractable one. Causal modeling can sometimes be a divide-and-conquer game, and various challenges that might initially seem impossible to tackle can be addressed (sometimes relatively easily) if we're open to refining our questions.

For instance, one mistake that I observe in the industry is starting with very broad questions regarding a *complete causal model* of a process, or even an entire organization/organizational unit. In certain cases, building such a *complete* model might be very difficult, very costly, or both.

Often, answering one or two well-defined causal questions can bring significant benefits at a fraction of the cost of such a big project.

The problem with *over-scoping* the project and asking too broad questions is not unique to causality. It happens in non-causal AI projects as well. That said, under-defined or incorrectly scoped questions can break a causal project much faster than a traditional machine learning project.

One positive is that an organization can understand relatively early on that a project does not bring the expected benefits and shut it down faster, avoiding significant losses. The negative is that such an experience might produce disappointment, leading to reluctance to move toward causal modeling. This can be detrimental because, for many organizations, causal modeling can bring significant benefits, which might often remain unrealized without causal methods.

A well-defined (set of) question(s) is vital to any successful causal project.

Obtaining expert knowledge

Expert knowledge is of paramount importance in causal modeling. Expert insights can help us disambiguate ambiguous edges in a causal graph, narrow down the search space, and evaluate the quality of the output of causal discovery algorithms. Many sources of information can be valuable here – domain knowledge, scientific literature, previous internal experiments, results of simulations, and so on.

Depending on the use case, collecting and validating expert knowledge can be a short and natural process or a long and effortful one, but its value is hard to overestimate.

Note that not all sources of expert knowledge have to come with the same level of confidence. It's a good idea to keep an open mind regarding the (less confident) sources and be prepared to discard them in the process, especially if alternative explanations are more plausible and/or more coherent with other trustworthy sources of information.

It's a good practice to store and manage expert knowledge in a structured and accessible way (e.g., following FAIR (<https://bit.ly/FAIRPrinciples>) or another set of principles that are well suited for your organization and the problem that you're solving).

This is of special importance in the case of larger systems. A lack of good access to collected expert knowledge can lead to frustration and consume a project team's creative energy, which should be directed at solving a problem rather than dealing with avoidable technical issues.

Once collected, information should be stored in a way that enables us to easily access it in the future. This way, we can make sure that we optimally use our resources if this particular knowledge can be reused in future projects.

After collecting expert knowledge, we're ready to encode it as a graph.

In reality, expert knowledge collection and graph generation can be intertwined, and multiple iterations between these two steps can occur.

Generating hypothetical graph(s)

After defining the question and collecting available expert knowledge, we are ready to generate the first hypothetical graph (or a set of hypothetical graphs).

These graphs represent our hypotheses regarding the data-generating process.

They don't have to be perfect or complete at this stage. Hypothetical graphs will allow us to understand what we know and whether we have enough information to answer the question(s) of interest.

Sometimes, even graphs that look non-sufficiently informative at first sight might contain enough information to answer some (or all) of our questions.

Depending on the size of the graph, the size of the project, and your organization characteristics, we might choose different storage options for our graphs – a repository containing (a) file(s) encoding the graph(s) or a graph database (e.g., Neo4j or Amazon Neptune) can all be valid options for you.

Accessibility and the ease of updating the structure are of key importance here.

Check identifiability

It's likely that your first graph will contain some ambiguous edges and/or some unobserved variables. The effect of interest might be possible to estimate even despite this.

To check whether this is the case, you can use one of the advanced identifiability algorithms. You can find some of these algorithms in the Python `grap1-causal` library, developed by Max Little of the University of Birmingham and MIT.

Check the repository and demo notebooks here: <https://bit.ly/GRAPLCausal>.

Even if your effect is not identifiable right away, you might still be able to obtain some actionable information from your model. This is possible for a broad array of models with sensitivity analysis tools (check *Chapter 8* for more details).

For instance, if you work in marketing or sales, you might know from experience that even if there's hidden confounding in your data, the maximum impact of all hidden confounders on sales should not be greater than some particular value.

If this is the case, you can reliably check whether your effect holds under extreme confounding or under the most likely values of confounders.

If your effect turns out to be identifiable right away or you work with experimental data, you can start estimating the effect using one of the methods that we discussed in *Part 2* of our book.

If you have doubts regarding some of the edges or their orientation, you can employ one of the causal discovery methods that we discussed in *Part 3* and confront the output against expert knowledge. If you can afford interventions on all or some variables, methods such as **ENCO** (*Chapter 14*) can lead to very good results.

Falsifying hypotheses

When we obtain an identifiable graph, we treat it as a hypothesis, and we can now learn the functional assignments over the graph (e.g., using the four-step framework from *Chapter 7*). After learning the functional assignments over the graph, we can generate predictions. Testing these predictions over interventional test data is a good way to check whether the model behaves realistically.

Strong discrepancies between predictions and actual effects are a strong indicator that a model has issues (structural, related to estimation or both). A lack of discrepancies does not automatically guarantee that the model is correct (recall the Popperian logic of falsification that we discussed in *Chapter 7* and *Chapter 12*).

The five steps (defining the question(s), collecting expert knowledge, generating a graph, checking identifiability, and falsifying the hypotheses) are usually performed iteratively, and sometimes, their ordering may be altered. They might also involve some advanced techniques that we did not discuss in our book – root cause analysis, causal outlier detection, generating counterfactuals from a functional model, and so on.

Note that the steps we described do not include the actual data collection process. This is because data collection might happen at very different stages, depending on the nature of the question, your organization's data maturity, and the methodology you pick.

In this section, we discussed five steps we can take in order to carry out a reliable causal project:

1. Starting with a question.
2. Obtaining expert knowledge.
3. Generating hypotheses.
4. Checking identifiability.
5. Falsifying the hypotheses.

In the next section, we'll see examples of causal projects carried out in the real world.

Causality and business

In this section, we'll describe a couple of real-world use cases where causal systems have been successfully implemented to address business challenges and discuss how causality intersects with business frameworks.

How causal doers go from vision to implementation

Geminos is a US-based company, with a mission to help businesses across industries solve their challenges using causality. One of Geminos' success stories that I particularly like comes from their engagement with an industrial manufacturer of metal products; let's call them Company M.

I like this story, because it emphasizes the broad applicability of causality that goes way beyond the most popular cases in marketing and econometrics.

Let's see how it worked.

Company M was interested in optimizing the process of production of one of its products. They formulated four main questions:

1. How do we minimize an important characteristic of product P ?
2. Which variables have the strongest effect on outcome O ?
3. What has caused a recent change in outcome O and how do we revert the change?
4. Why do outliers occur and how do we remove them?

Note that all four questions require at least interventional (rung 2) reasoning, which makes them unanswerable using traditional statistical or machine learning methods.

For instance, question 1 asks what *actions* should be taken to minimize the characteristic of interest of product *P*. This question goes beyond plain understanding of what simply correlates with this characteristic (for context, recall our discussion on confounding from *Chapter 1*). To answer question 1, we need to understand the *mechanism* behind this characteristic and how changes in other variables can affect it.

After defining the questions, Geminos' team consulted the experts and researched relevant scientific literature. Combining information from both sources, the team built their first hypothetical graph.

Company M had recorded a rich set of variables describing their processes for a couple of years before the project started, yet it turned out that some of the variables considered in the graph had not been collected up to this point.

The team addressed this challenge by adjusting the graph in a way that preserved identifiability of the most important relationships, and the client learned which variables should be prioritized for future collection to facilitate further causal analyses.

The knowledge about which variables' collection to prioritize in the future was valuable in itself, as it helped the client understand how to optimally allocate resources. Installing new sensors or altering the existing software architecture can be very expensive, and when it does not lead to insights that bring real business value, it can cause significant losses.

With the updated graph, the team was ready to start validating their hypotheses. They estimated the coefficients in the model described by the hypothetical graph using Geminos' DoWhy-powered platform.

The team compared model outcomes against observational and interventional data and iteratively improved the model. As Owen from Geminos emphasized, "*Interrogating causal models is an important element of the iterative approach taken to causal AI.*"

After a number of iterations, the client was able to answer their questions and optimize the process beyond what they had previously been able to achieve, using traditional machine learning, statistical and optimization tools.

Company M spotted the outliers using causal outlier detection and understood the causal chain behind recent changes, ultimately answering all four key questions they formulated at the beginning of the project. This led to significant improvements in the production process.

The company also learned how to structure and prioritize their new data collection efforts in order to maximize the value, which saved them potential losses related to adapting existing systems to collect data that would not bring clear business value.

Geminos is not the only organization that successfully implements and productionalizes causal solutions. A UK-based company, **causaLens**, offers a causal data science platform that abstracts much of the complexity of building causal models. causaLens has helped numerous customers across industries

build more robust solutions by introducing causality. Their use cases include marketing, supply chain, manufacturing optimization and more.

Causal modeling is also used in the digital entertainment industry. **Playtika**, an Israel-based digital entertainment company specializing in mobile gaming uses uplift (CATE) modeling to optimize their user experience. The company has recently open sourced their uplift evaluation library, `uplift-analysis` (check out <https://bit.ly/PlaytikaUplift>).

Causal modeling is also used by Swedish audio streaming giant **Spotify**. The company regularly shares their experience in applied causality through their technical blog and top conference publications (e.g., Jeunen et al., 2022). Spotify's blog covers a wide variety of topics, from sensitivity analysis of a synthetic control estimator (which we learned about in *Chapter 11*; see more at <https://bit.ly/SpotifySynthControl>) to disentangling causal effects from sets of interventions under hidden confounding (<https://bit.ly/SpotifyHiddenBlog>).

Production, sales, marketing and digital entertainment are not the only areas that can benefit from causal modeling. Causal models are researched and implemented across fields, from medicine to the automotive industry.

This is not surprising, as causal models often capture the aspects of problems that we're the most interested in solving. Causal models (in particular **structural causal models (SCMs)**) are also compatible with (or easily adaptable to) numerous business and process improvement frameworks.

Let's see an example. **Six Sigma** is a set of process improvement tools and techniques proposed by Bill Smith at Motorola. Within the Six Sigma framework, causality is defined as an impact of some input x on some output y via some mechanism f , formally:

$$y := f(x)$$

Note that this is almost identical to functional assignments in SCMs, which we discussed back in *Chapter 2* (except that we skip the noise variable in the preceding formula).

This makes causal analysis a natural choice whenever decision-making questions are at play.

To learn more about causality from a business point of view, check out *Causal Artificial Intelligence: The Next Step in Effective, Efficient, and Practical AI* by Judith Hurwitz and John K. Thompson (due for release in Q4 2023). Now, let's take a look at the future of causality.

Toward the future of causal ML

In this section, we'll briefly explore some possible future directions for causality from business, application, and research point of views. As always when talking about the future, this is somewhat of a gamble, especially in the second part of this section where we will discuss more advanced ideas.

Let's start our journey into the future from where we're currently standing.

Where are we now and where are we heading?

With an average of 3.2 new papers published on arXiv every day in 2022, causal inference has exploded in popularity, attracting a large amount of talent and interest from top researchers and institutions, including industry giants such as Amazon or Microsoft.

At the same time, for many organizations, causal methods are much less accessible than traditional statistical and machine learning techniques. This state of affairs is likely driven by a strong focus of educational system on associational methods when teaching about data science and machine learning, along with a lack of accessible materials that combine the theoretical and practical aspects of causality (I hope that this book will be a small step in changing the latter).

A number of aspects of causal modeling and causal thinking can be relatively easily adopted by many organizations, bringing them strong benefits.

Here are three skills that offer significant benefits for a wide range of organizations:

- An awareness of the differences between the rungs of the *Ladder of Causation* can help analysts clearly distinguish between problems that can be solved using traditional associational methods and ones that cannot be solved this way. Such awareness empowers organizations to mitigate losses related to investing in machine learning projects that answer ill-posed questions and cannot succeed in the long run.
- The ability to think structurally about a data-generating process and to understand conditional independence structure can help analysts, data scientists, and engineers make better decisions regarding statistical control and understand the challenges to the robustness of their models under distribution shifts. This ability enables teams across an organization to avoid costly surprises caused by misleading predictions generated by mis-specified models.
- The ability to model **conditional average treatment effects (CATE)** can help uncover information hidden by traditional A/B testing analysis techniques and bring new opportunities to personalize and revolutionize user experience. Causal personalization might bring better returns than traditional recommender systems. This is true in marketing, churn prevention, and other areas, thanks to CATE models' capability to recognize which units should be treated and which ones should remain untreated.

Despite the relatively low entry barrier, these three ideas seem to be highly underutilized across sectors, industries, and organization types.

Causal benchmarks

From a research perspective, one of the main challenges that we face today in causality is a lack of widely accessible real-world datasets and universal benchmarks – the analogs of ImageNet in computer vision.

Datasets and benchmarks can play a crucial role in advancing the field of causality by fostering reproducibility and transparency and providing researchers with a common point of reference.

Today's synthetic benchmarks have properties that hinder their effective use in causal discovery (Reisach et al., 2021) and causal inference (Curth et al., 2021).

The topic of causal benchmarks is actively researched, and initiatives such as CLear 2023's *Call for Causal Datasets* (<https://bit.ly/CLearDatasets>) demonstrate the community's rising awareness and readiness to tackle these challenges.

Some useful data simulators that can help to benchmark causal discovery methods have been also proposed recently (see <https://bit.ly/DiagnosisSimulator>).

Now, let's discuss four potential research and application directions where causality can bring value.

Causal data fusion

Causal data fusion (Bareinboim & Pearl, 2016, and Hünermund & Bareinboim, 2023) is an umbrella term for combining data from different sources to perform causal inference. In particular, interventional data might be combined with observational data. Causal data fusion might be particularly useful when combined with uncertainty estimation (e.g., Chau et al., 2021) and active learning paradigms (e.g., Toth et al., 2022).

Causal data fusion's promise is to make causal inference efficient by leveraging information from multiple datasets coming from different sources. This is particularly beneficial in biomedical sciences, where experiments might be expensive, difficult, or risky. Causal data fusion can help make valid and trustworthy conclusions by combining the information from small-sample experiments and large-sample observational datasets.

Although a number of challenges in causal data fusion are effectively solved (Bareinboim & Pearl, 2016), the paradigm seems underutilized in practice, in particular in industry.

This could be perhaps changed by making stakeholders more familiar with the opportunities that causal data fusion offers.

Intervening agents

The 2022/23 generative AI revolution has resulted in a number of new applications, including Auto-GPT or AgentGPT – programs that leverage GPT-class models behind the scenes and allow them to interact with the environment.

Model instances (called *agents*) might have access to the internet or other resources and can solve complex multistep tasks. Equipping these agents with causal reasoning capabilities can make them much more effective and less susceptible to confounding, especially if they are able to interact with the environment in order to falsify their own hypotheses about causal mechanisms.

Agents such as these could perform automated research and are a significant step toward creating the *automated scientist*. Note that allowing these agents to interact with the physical environment rather than only with the virtual one could significantly enhance their efficiency, yet this also comes with a number of important safety and ethical considerations.

Note that these ideas are related to some of the concepts discussed extensively by Elias Bareinboim in his 2020 *Causal Reinforcement Learning* tutorial (<https://bit.ly/CausalRL>).

Causal structure learning

Causal structure learning (Schölkopf et al., 2021) is a broad term encompassing various algorithms aimed at decoding a causal structure from high-dimensional data (e.g., video). Contrary to traditional causal discovery, causal structure learning assumes that some lower-dimensional latent variables exist that describe the causal structure of a system presented in the original high-dimensional representation.

For example, CITIRS (Lippe et al., 2022) learns low-level causal structures from videos containing interventions.

Depending on a model, causal structure models can work in various scenarios (e.g., with known or unknown interventions). This line of research can help us build systems that can intervene in the world and learn causal structures by observing the effects of their interventions (using vision or other modalities).

Note that the term *causal structure learning* is sometimes used interchangeably with the term *causal discovery*, and sometimes *causal structure learning* is understood as a superset of *causal discovery*, which includes the models with and without latent variables.

Imitation learning

When human babies learn basic world models, they often rely on experimentation (e.g., Gopnik, 2009; also vide *Chapter 1* of this book). In parallel, children imitate other people in their environment. At later stages in life, we start learning from others more often than performing our own experiments. This type of learning is efficient but might be susceptible to arbitrary biases or confounding.

Racial or gender stereotypes passed through generations are great examples of learning incorrect world models by imitation. For instance, in modern Western culture, it was widely believed until the late 1960s that women are *physically incapable* of running a marathon. Bobbi Gibb falsified this hypothesis by (illegally) running in the Boston Marathon in 1966. She not only finished the race but also ran faster than roughly 66% of the male participants (<https://bit.ly/BobbiGibbStory>).

Imitation learning is often more sample-efficient than experiments but does not guarantee causal identifiability. We face a similar challenge in modern **natural language processing (NLP)**. Some **large language models (LLMs)** offer powerful causal capabilities learned from language data, but they sometimes fail unpredictably (Kıcıman et al., 2023).

Willig et al. (2023) proposed that LLMs learn a **meta-SCM** from text and that this model is associational in its nature.

Agents learning by imitation and then falsifying learned models by using interventions, or carrying out efficient causal reasoning by leveraging causal data fusion, can be great and efficient learners, capable of fast adaptation to various domains.

For instance, causally aware imitation learning can be very useful to create virtual assistants for business, medicine, or research. These assistants can learn from existing materials or the performance of other (virtual or physical) assistants and then improve by validating the trustworthiness of learned models.

Another broad research direction that intersects with some of the preceding ideas is neuro-symbolic AI – a class of models combining associational representation learning with symbolic reasoning modules.

The causal analysis of LLMs that we mentioned earlier in this section is in itself a promising research path that brings many interesting questions to the table. Thanks to the broad adoption of LLMs, this path has a chance to spark broader interest and attract the funding necessary to achieve significant progress in the near future.

Learning causality

In this section, we'll point to the resources to learn more about causality after finishing this book.

For many people starting with causality, their learning path begins with excitement. The promise of causality is attractive and powerful. After learning about the basics and realizing the challenges that any student of causality has to face, many of us lose hope in the early stages of our journeys.

Some of us regain it, learning that solutions do exist, although not necessarily where we initially expected to find them.

After overcoming the first challenges and going deeper into the topic, many of us realize that there are more difficulties to come. Learning from earlier experiences, it's easier at this stage to realize that (many of) these difficulties can be tackled using a creative and systematic approach.

I like the way the Swiss educator and researcher Quentin Gallea presented the journey into learning causality in a graphical form (*Figure 15.1*).

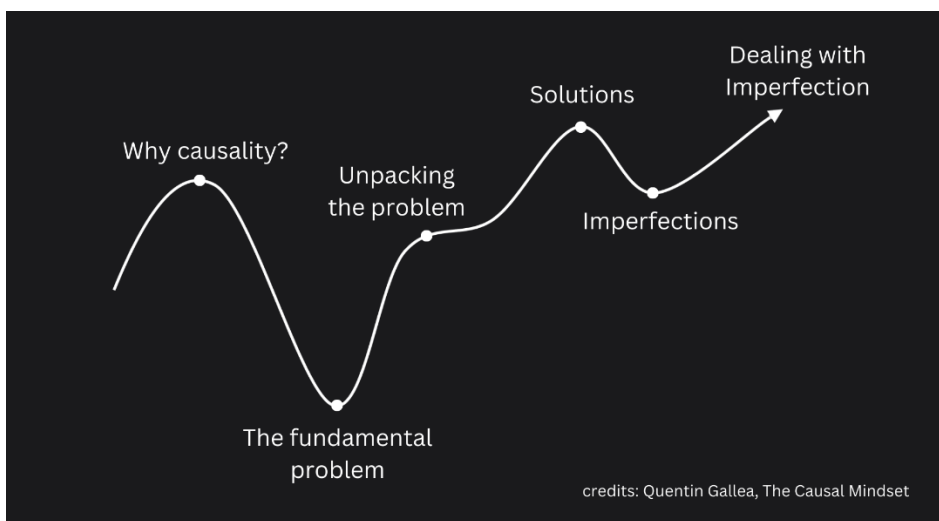


Figure 15.1 – The journey into causality by Quentin Gallea

The figure comes from *The Causal Mindset* (<https://bit.ly/QuentinGallea>), a book that Quentin is working on.

At whichever point of the curve from *Figure 15.1* you find yourself currently, being consistent will inevitably move you to the next stage.

A common problem that many of us face when learning a new topic is the choice of the next resource after finishing a book or a course.

Here are a couple of resources that you might find useful on your journey one day.

First, there are many great books on causality. Starting with Judea Pearl's classics such as *The Book of Why* and finishing with Hernán and Robins' *What If?*, you can learn a lot about different perspectives on causal inference and discovery. I summarized six great books on causality in one of my blog posts here: <https://bit.ly/SixBooksBlog>.

Second, survey papers are a great way to get a grasp of what's going on in the field and what the open challenges are.

Here are three survey papers that can help you understand the current causal landscape:

- *Causal machine learning: A survey and open problems* (Kadour et al., 2022)
- *Deep Causal Learning: Representation, Discovery and Inference* (Deng et al., 2022)
- *D'ya like dags? A survey on structure learning and causal discovery* (Vowels et al., 2022)

Additionally, for a unifying perspective on various causal methods, check out the *Causal Deep Learning* paper by Jeroen Berrevoets and colleagues (Berrevoets et al., 2023).

To stay up to date, get free learning resources, and become a part of a community of like-minded causal learners, subscribe to my free weekly *Causal Python* newsletter: <https://bit.ly/CausalPython>

Let's stay in touch

Community is a catalyst for growth. Let's connect on LinkedIn and Twitter so that we can learn from each other:

- LinkedIn: <https://www.linkedin.com/in/aleksandermolak/>
- Twitter: @AleksanderMolak

If you want to consult a project or run a workshop on causality for your team, drop me a line at alex@causalpython.io.

For comments and questions regarding this book, email me at book@causalpython.io.

Wrapping it up

It's time to conclude our journey.

In this chapter, we summarized what we've learned in this book, discussed five steps to make the best out of our causal projects, took a look at the intersection of causality and business, and sneaked into the (potential) future of causal research and applications. Finally, we listed a number of resources that you might find useful in the next stages of your causal journey.

I hope finishing this book won't be the end for you, but rather the beginning of a new causal chapter!

I hope to see you again!

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